

Generalized Vibrational Analysis

Notation

q^μ	Cartesian displacement coordinate, $q^\mu = x^\mu - x_0^\mu$
g_μ	Cartesian gradient, $g_\mu = \frac{\partial V}{\partial q^\mu} = \frac{\partial V}{\partial x^\mu}$
$H_{\mu\nu}$	Cartesian Hessian, $H_{\mu\nu} = \frac{\partial^2 V}{\partial q^\mu \partial q^\nu} = \frac{\partial^2 V}{\partial x^\mu \partial x^\nu}$
$\mathbf{v}, \mathbf{w}$	Cartesian vectors with components v^μ, w^μ
α, β	Cartesian co-vectors with components α_μ, β_μ
$\mathbf{v}^\flat$	Dual (lowered) Cartesian co-vector with components $v_\mu = G_{\mu\nu} v^\nu$
$\alpha^\sharp$	Dual (raised) Cartesian vector with components $\alpha^\mu = G^{\mu\nu} \alpha_\nu$
$G_{\mu\nu}$ and $G^{\mu\nu}$	Co- and contravariant Cartesian coordinate metrics
$\langle \mathbf{v}, \mathbf{w} \rangle$	Cartesian vector inner product, $\langle \mathbf{v}, \mathbf{w} \rangle = G_{\mu\nu} v^\mu w^\nu$
$\langle \alpha, \beta \rangle$	Cartesian co-vector inner product, $\langle \alpha, \beta \rangle = G^{\mu\nu} \alpha_\mu \beta_\nu$
$\tilde{q}^\mu$	Mass-weighted Cartesian displacement coordinate, $\tilde{q}^\mu = m_{(\mu)}^{\frac{1}{2}} q^\mu$
$\tilde{g}_\mu$	Mass-weighted Cartesian gradient, $\tilde{g}_\mu = \frac{\partial V}{\partial \tilde{q}^\mu} = m_{(\mu)}^{-\frac{1}{2}} \frac{\partial V}{\partial q^\mu}$
$\tilde{H}_{\mu\nu}$	Mass-weighted Cartesian Hessian, $\tilde{H}_{\mu\nu} = \frac{\partial^2 V}{\partial \tilde{q}^\mu \partial \tilde{q}^\nu} = m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} \frac{\partial^2 V}{\partial q^\mu \partial q^\nu}$
$\tilde{\mathbf{v}}, \tilde{\mathbf{w}}$	Mass-weighted Cartesian vectors with components $\tilde{v}^\mu, \tilde{w}^\mu$
$\tilde{\alpha}, \tilde{\beta}$	Cartesian co-vectors with components $\tilde{\alpha}_\mu, \tilde{\beta}_\mu$
$\tilde{G}_{\mu\nu}$ and $\tilde{G}^{\mu\nu}$	Co- and contravariant mass-weighted Cartesian coordinate metrics
$\langle \tilde{\mathbf{v}}, \tilde{\mathbf{w}} \rangle$	Cartesian vector inner product, $\langle \tilde{\mathbf{v}}, \tilde{\mathbf{w}} \rangle = \tilde{G}_{\mu\nu} \tilde{v}^\mu \tilde{w}^\nu$
$\langle \tilde{\alpha}, \tilde{\beta} \rangle$	Cartesian co-vector inner product, $\langle \tilde{\alpha}, \tilde{\beta} \rangle = \tilde{G}^{\mu\nu} \tilde{\alpha}_\mu \tilde{\beta}_\nu$
q^i	Internal displacement coordinate
g_i	Internal coordinate gradient, $g_i = \frac{\partial V}{\partial q^i}$
H_{ij}	Internal coordinate Hessian, $H_{ij} = \frac{\partial^2 V}{\partial q^i \partial q^j}$
$\mathbf{v}', \mathbf{w}'$	Internal coordinate vectors with components v^i, w^i
α', β'	Internal coordinate co-vectors with components α_i, β_i
G_{ij} and G^{ij}	Co- and contravariant Cartesian coordinate metrics
$\langle \mathbf{v}', \mathbf{w}' \rangle$	Internal coordinate vector inner product, $\langle \mathbf{v}', \mathbf{w}' \rangle = G_{ij} v^i w^j$
$\langle \alpha', \beta' \rangle$	Internal coordinate co-vector inner product, $\langle \alpha', \beta' \rangle = G^{ij} \alpha_i \beta_j$
B_μ^i	Internal coordinate Jacobian with respect to Cartesians, $B_\mu^i \equiv \frac{\partial q^i}{\partial q^\mu}$
$\tilde{B}_\mu^i$	Internal coordinate Jacobian with respect to mass-weighted Cartesians, $\tilde{B}_\mu^i \equiv \frac{\partial q^i}{\partial \tilde{q}^\mu}$
$\tilde{C}_{\mu\nu}^i$	Derivative of the mass-weighted internal coordinate Jacobian, $C_{\mu\nu}^i = \frac{\partial \tilde{B}_\mu^i}{\partial \tilde{q}^\nu} \equiv \frac{\partial^2 q^i}{\partial \tilde{q}^\mu \partial \tilde{q}^\nu}$

Proofs

Proposition 1. Contravariance of coordinates.¹

$$e_i = e_\mu(T)_i^\mu \Rightarrow q^i = (T^{-1})_i^\mu q^\mu$$

By definition, this behavior holds for all vectors and for all contravariant (upper index) tensor components.

Proposition 2. Covariance of derivatives.²

$$e_i = e_\mu(T)_i^\mu \Rightarrow \frac{\partial f}{\partial q^i} = \frac{\partial f}{\partial q^\mu}(T)_i^\mu$$

By definition, this behavior holds for all co-vectors and for all covariant (lower index) tensor components.

Proposition 3. Invariance of inner products.³

$$\langle v, w \rangle = \langle \tilde{v}, \tilde{w} \rangle = \langle v', w' \rangle \quad \langle \alpha, \beta \rangle = \langle \tilde{\alpha}, \tilde{\beta} \rangle = \langle \alpha', \beta' \rangle$$

Since the vector and co-vector inner products are equivalent, we can simply write $\langle v, w \rangle$ and $\langle \alpha, \beta \rangle$.

Proposition 4. Co- and contravariant metrics.⁴ The co- and contravariant metrics are inverses.

$$G_{jk}G^{ki} = G^{ik}G_{kj} = \delta_j^i$$

Proposition 5. Kinetic energy metrics.⁵ Defining $T = \frac{1}{2}\langle \dot{q}, \dot{q} \rangle$ yields the following system of metrics.

$$G_{\mu\nu} = m_{(\mu)}\delta_{\mu\nu} \quad G^{\mu\nu} = m_{(\mu)}^{-1}\delta^{\mu\nu} \quad \tilde{G}_{\mu\nu} = \delta_{\mu\nu} \quad \tilde{G}^{\mu\nu} = \delta^{\mu\nu} \quad G_{ij} = \text{inverse of } G^{ij} = \delta^{\mu\nu}\tilde{B}_\mu^i\tilde{B}_\nu^j$$

In the last case, the given internal coordinate metric is only exact for infinitesimal displacements, dq^i .

Proposition 6. Internal coordinate Taylor expansion in mass-weighted Cartesian coordinates.⁶

$$q^i = \tilde{B}_\mu^i \tilde{q}^\mu + \tilde{C}_{\mu\nu}^i \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^2)$$

¹Proof: $q^\mu e_\mu = \mathbf{q} = q^i e_i = q^i e_\mu(T)_i^\mu = (T)_i^\mu q^i e_\mu$. This implies $q^\mu = (T)_i^\mu q^i$, which leads to the final result: $(T^{-1})_i^\mu q^\mu = (T^{-1})_i^\mu (T)_j^\nu q^j = \delta_j^i q^j = q^i$.

²Proof: In Proposition 1, it was proven that $q^\mu = (T)_i^\mu q^i$. The final result then follows from the chain rule: $\frac{\partial f}{\partial q^\mu} = \frac{\partial f}{\partial q^i} \frac{\partial q^i}{\partial q^\mu} = \frac{\partial f}{\partial q^i} (T)_j^\mu \frac{\partial q^j}{\partial q^\mu} = \frac{\partial f}{\partial q^\mu} (T)_j^\mu \delta_j^i = \frac{\partial f}{\partial q^\mu} (T)_i^\mu$.

³Proof: Given a basis transformation $e_i = e_\mu(T)_i^\mu$, covariant quantities satisfy $\alpha_i = \alpha_\mu(T)_i^\mu$ and contravariant quantities satisfy $v^\mu = v^i(T)_i^\mu$. This implies $\langle v', w' \rangle = G_{ij}v^i w^j = G_{\mu\nu}(T)_i^\mu (T)_j^\nu v^i w^j = G_{\mu\nu}v^\mu w^\nu = \langle v, w \rangle$. The same logic holds for all of the other cases.

⁴Proof: Let the e_i basis be derived from an orthonormal basis e_μ as $e_i = e_\mu(T)_i^\mu$. This is always possible by Gram-Schmidt orthogonalization with respect to the final metric G_{ij} . It follows that $G_{\mu\nu} = \delta_{\mu\nu}$ and $G^{\mu\nu} = \delta^{\mu\nu}$. This leads to the final result: $G_{jk}G^{ki} = G_{\nu\rho}G^{\rho\mu}(T)_j^\nu (T^{-1})_i^\mu = \delta_{\nu\rho}G^{\rho\mu}(T)_j^\nu (T^{-1})_i^\mu = (T)_j^\mu (T^{-1})_i^\mu = \delta_j^i$.

⁵Proof: For the Cartesian metric, $\frac{1}{2}\langle \dot{q}, \dot{q} \rangle = \frac{1}{2}G_{\mu\nu}\dot{q}^\mu \dot{q}^\nu = \frac{1}{2}m_{(\mu)}\delta_{\mu\nu}\dot{q}^\mu \dot{q}^\nu = T$. By Proposition 4, the corresponding contravariant metric is $G^{\mu\nu} = m_{(\mu)}^{-1}\delta^{\mu\nu}$. By Proposition 3, the proof is completed by showing that this transforms into the other contravariant metrics. The transformation to mass-weighted Cartesians is $\tilde{q}^\mu = m_{(\mu)}^{\frac{1}{2}}q^\mu$, so the corresponding contravariant metric in mass-weighted coordinates is $\tilde{G}^{\mu\nu} = m_{(\mu)}^{\frac{1}{2}}m_{(\nu)}^{\frac{1}{2}}G^{\mu\nu} = m_{(\mu)}^{\frac{1}{2}}m_{(\nu)}^{\frac{1}{2}}m_{(\mu)}^{-1}\delta^{\mu\nu} = m_{(\nu)}^{\frac{1}{2}}m_{(\mu)}^{-\frac{1}{2}}\delta^{\mu\nu} = \delta^{\mu\nu}$. The transformation from mass-weighted to internal coordinates is $q^i = \tilde{B}_\mu^i \tilde{q}^\mu + \mathcal{O}(\tilde{q}^2)$, so the corresponding contravariant metric in internal coordinates is $G^{ij} = \tilde{G}^{\mu\nu}\tilde{B}_\mu^i \tilde{B}_\nu^j = \delta^{\mu\nu}\tilde{B}_\mu^i \tilde{B}_\nu^j$.

⁶Proof: This follows from writing the Taylor expansion of q^i in terms of $\tilde{B}_\mu^i$ and $\tilde{C}_{\mu\nu}^i$, as defined above.

Proposition 7. Right pseudo-inverse of the B-matrix.⁷

$$\tilde{B}_\rho^i \tilde{A}_j^\rho = \delta_j^i \quad \tilde{A}_i^\mu \equiv \delta^{\mu\rho} \tilde{B}_\rho^k G_{ki}$$

Proposition 8. Potential energy derivatives.⁸ The potential energy $V = V_0 + g_\mu q^\mu + \frac{1}{2} H_{\mu\nu} q^\mu q^\nu + \mathcal{O}(q^2)$ can be expressed in the other coordinate systems by the following gradient and Hessian transformations.

$$\begin{aligned} V &= V_0 + \tilde{g}_\mu \tilde{q}^\mu + \frac{1}{2} \tilde{H}_{\mu\nu} \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^2) & \tilde{g}_\mu &= m_{(\mu)}^{-\frac{1}{2}} g_\mu & \tilde{H}_{\mu\nu} &= m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} H_{\mu\nu} \\ V &= V_0 + g_i q^i + \frac{1}{2} H_{ij} q^i q^j + \mathcal{O}(q^2) & g_i &= \tilde{g}_\mu \tilde{A}_i^\mu & H_{ij} &= (\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k) \tilde{A}_i^\mu \tilde{A}_j^\nu \end{aligned}$$

Proposition 9. Dual vector contraction as inner product.⁹

$$v^i w_i = \langle v, w \rangle \quad \alpha_i \beta^i = \langle \alpha, \beta \rangle$$

Proposition 10. Dual vector consistency.^{10 11}

$$(v^\flat)^\sharp = v \quad (\alpha^\sharp)^\flat = \alpha$$

Proposition 11. Projection operator.¹² The projections onto v and α are as follows.

$$P_v = \frac{v \otimes v^\flat}{\langle v, v \rangle} \quad (P_v)_j^i = \frac{v^i v_j}{\langle v, v \rangle} \quad P_\alpha = \frac{\alpha \otimes \alpha^\sharp}{\langle \alpha, \alpha \rangle} \quad (P_\alpha)_j^i = \frac{\alpha_j \alpha^i}{\langle \alpha, \alpha \rangle}$$

⁷ Proof: By Proposition 16, $\delta_j^i = G^{ik} G_{kj} = \delta^{\rho\sigma} \tilde{B}_\rho^i \tilde{B}_\sigma^k G_{kj} = \tilde{B}_\rho^i \tilde{A}_j^\rho$.

⁸ Proof: Because mass-weighting is a linear transformation, the mass-weighted Cartesian gradient and Hessian can be obtained directly using $\tilde{\alpha}_\mu = m_{(\mu)}^{-\frac{1}{2}} \alpha_\mu$. The transformation to internal coordinates is non-linear and requires further consideration. First, substituting Proposition 6 into the internal coordinate Taylor expansion and collecting terms gives $V = g_i \tilde{B}_\mu^i \tilde{q}^\mu + \frac{1}{2} (g_i \tilde{C}_{\mu\nu}^i + H_{ij} \tilde{B}_\mu^i \tilde{B}_\nu^j) \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^2)$. Comparing this to the mass-weighted Taylor expansion allows us to identify $\tilde{g}_\mu = g_i \tilde{B}_\mu^i$ and $\tilde{H}_{\mu\nu} = g_i \tilde{C}_{\mu\nu}^i + H_{ij} \tilde{B}_\mu^i \tilde{B}_\nu^j$. Inverting these by Proposition 7 gives the result.

⁹ Proof: By the definition of the dual vectors, $\langle v, w \rangle = G_{ij} v^i w^j = v^i w_i$ and $\langle \alpha, \beta \rangle = G^{ij} \alpha_i \beta_j = \alpha_i \beta^i$.

¹⁰ Proof: By Proposition 4, $G^{ij} G_{jk} v^k = \delta_k^i v^k = v^i$ and $G_{ij} G^{jk} \alpha_k = \delta_i^k \alpha_k = \alpha_i$.

¹¹ An alternative approach takes this property as fundamental and derives Proposition 4 from it.

¹² Proof: By Proposition 9, $(P_v(w))^i = \frac{v^i v_j}{\langle v, v \rangle} w^j = v^i \frac{\langle v, w \rangle}{\langle v, v \rangle} \Rightarrow P_v(w) = v \frac{\langle v, w \rangle}{\langle v, v \rangle}$ and $(P_\alpha(\beta))_j = \frac{\alpha_j \alpha^i}{\langle \alpha, \alpha \rangle} \beta_i = \alpha_j \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \Rightarrow P_\alpha(\beta) = \alpha \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}$. This is the standard form of a standard projection operator, albeit with a generalized inner product.

Proposition 12. Lemma: Internal to mass-weighted Cartesian coordinate co-vector transformation.¹³

$$\tilde{v}_\mu = v_k \tilde{B}_\mu^k$$

Proposition 13. Kinetic Energy Metrics: Consistency.¹⁴ If $\langle \cdot, \cdot \rangle_m$ is chosen as the inner product for Cartesian coordinates, then $\langle \cdot, \cdot \rangle_1$ is the corresponding inner product for mass-weighted Cartesian coordinates and $\langle \cdot, \cdot \rangle_G$ is the corresponding inner products for internal coordinates.

Proposition 14. Kinetic Energy Metrics.¹⁵

$$T = \frac{1}{2} \langle \dot{x}, \dot{x} \rangle_m = \frac{1}{2} \langle \tilde{x}, \tilde{x} \rangle_1 = \frac{1}{2} \langle \dot{q}, \dot{q} \rangle_G$$

Proposition 15. Lemma: Cartesian Gradient and Hessian Taylor Expansion.¹⁶

$$\tilde{g}_\mu = g_k \tilde{B}_\mu^k \quad \tilde{H}_{\mu\nu} = H_{kl} \tilde{B}_\mu^k \tilde{B}_\nu^l + g_k \frac{\partial^2 q^k}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu}$$

Proposition 16. Internal Coordinate Metrics.¹⁷ The covariant and contravariant internal coordinate metric can be defined relative to mas-weighted Cartesians as follows.

$$G_{jk} G^{ki} = G^{ik} G_{kj} = \delta_j^i \quad G^{ij} \equiv \delta^{\rho\sigma} \tilde{B}_\rho^i \tilde{B}_\sigma^j$$

Proposition 17. Lemma: Jacobian Right Pseudo-inverse.¹⁸

$$\tilde{B}_\rho^i \tilde{A}_j^\rho = \delta_j^i \quad \tilde{A}_i^\mu \equiv \delta^{\mu\rho} \tilde{B}_\rho^k G_{ki}$$

Proposition 18. Internal Coordinate Gradient and Hessian.¹⁹

$$g_i = \tilde{g}_\rho \tilde{A}_i^\rho \quad H_{ij} = \left(\tilde{H}_{\rho\sigma} - g_k \frac{\partial^2 q^k}{\partial \tilde{x}^\rho \partial \tilde{x}^\sigma} \right) \tilde{A}_i^\rho \tilde{A}_j^\sigma$$

¹³Proof: The gradient $\tilde{g}_\mu = \frac{\partial V}{\partial \tilde{x}^\mu}$ is a co-vector, so all other co-vectors (and all lower indices) must transform like it. We can then prove this result by applying chain rule to the gradient: $\frac{\partial V}{\partial \tilde{x}^\mu} = \frac{\partial V}{\partial q^k} \frac{\partial q^k}{\partial \tilde{x}^\mu}$.

¹⁴Proof: First, consider the mass-weighted Cartesian co-vector inner product $\langle \tilde{\alpha}, \tilde{\beta} \rangle_1 = \tilde{\alpha}_\mu \tilde{\beta}_\nu \delta^{\mu\nu} = \alpha^\mu \beta^\nu m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} \delta^{\mu\nu} = \alpha^\mu \beta^\nu m_{(\mu)} \delta^{\mu\nu} = \alpha^\mu \beta^\nu m_{\mu\nu} = \langle \alpha, \beta \rangle_m$. Second, consider the internal coordinate co-vector inner product $\langle \alpha', \beta' \rangle_G =$

¹⁵Proof: The first expression is simply the definition of the kinetic energy, $T = \frac{1}{2} m_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$. The other expressions then follow from the behavior of the metric under coordinate transformation.

¹⁶Proof: The potential can be expressed in mass-weighted Cartesian coordinates as $V = V_0 + \tilde{g}_\mu \tilde{x}^\mu + \frac{1}{2} \tilde{H}_{\mu\nu} \tilde{x}^\mu \tilde{x}^\nu + \mathcal{O}(\tilde{x}^2)$ or in internal coordinates as $V = V_0 + g_i q^i + \frac{1}{2} H_{ij} q^i q^j + \mathcal{O}(q^2)$. Furthermore, the internal coordinates can be expanded in terms of mass-weighted Cartesian coordinates as $q^i = \frac{\partial q^i}{\partial \tilde{x}^\mu} \tilde{x}^\mu + \frac{1}{2} \frac{\partial^2 q^i}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \tilde{x}^\mu \tilde{x}^\nu + \mathcal{O}(\tilde{x}^2)$. Substituting this into the second potential expression and comparing to the first gives the result.

¹⁷Proof: Since Cartesian coordinates are rectilinear, their contravariant metric is $G^{\rho\sigma} = \delta^{\rho\sigma}$. The formula for the contravariant internal coordinate metric therefore follows from Proposition 20, and according to Proposition 19 the covariant internal coordinate metric is then defined as the inverse of this.

¹⁸Proof: By Proposition 16, $\delta_j^i = G^{ik} G_{kj} = \delta^{\rho\sigma} \tilde{B}_\rho^i \tilde{B}_\sigma^k G_{kj} = \tilde{B}_\rho^i \tilde{A}_j^\rho$.

¹⁹Proof: From Proposition 15 and Proposition 17, we have $\tilde{g}_\rho \tilde{A}_i^\rho = g_k \tilde{B}_\rho^k \tilde{A}_i^\rho = g_k \delta_i^k = g_i$ for the gradient and $\tilde{H}_{\rho\sigma} \tilde{A}_i^\rho \tilde{A}_j^\sigma = H_{kl} \tilde{B}_\rho^k \tilde{B}_\sigma^l \tilde{A}_i^\rho \tilde{A}_j^\sigma + g_k \frac{\partial^2 q^k}{\partial \tilde{x}^\rho \partial \tilde{x}^\sigma} \tilde{A}_i^\rho \tilde{A}_j^\sigma = H_{kl} \delta_i^k \delta_j^l + g_k \frac{\partial^2 q^k}{\partial \tilde{x}^\rho \partial \tilde{x}^\sigma} \tilde{A}_i^\rho \tilde{A}_j^\sigma = H_{ij} + g_k \frac{\partial^2 q^k}{\partial \tilde{x}^\rho \partial \tilde{x}^\sigma} \tilde{A}_i^\rho \tilde{A}_j^\sigma$.

General Results

Definitions.

$v_\mu \equiv g_{\mu\rho} v^\rho$ The “lowering” operation, which defines the dual of a vector

$v^\mu \equiv g^{\mu\rho} v_\rho$ This “raising” operation, which defines the contravariant metric

Proposition 19. Co- and Contravariant Metrics are Inverses.²⁰

$$g^{\mu\rho} g_{\rho\nu} = g_{\nu\rho} g^{\rho\mu} = \delta_\nu^\mu$$

Proposition 20. Metric Tensor Translation.²¹ The translation of a metric tensor from coordinate system x^μ to coordinate system x'^μ is given by the following.

$$g_{\mu\nu} = g'_{\rho\sigma} \frac{\partial x'^\rho}{\partial x^\mu} \frac{\partial x'^\sigma}{\partial x^\nu}$$

²⁰Proof: Lowering and raising an arbitrary vector in sequence gives $v^\mu = g^{\mu\rho} v_\rho = g^{\mu\rho} g_{\rho\nu} v^\nu$. This can only be satisfied if $g^{\mu\rho} g_{\rho\nu} = \delta_\nu^\mu$. Doing the same starting from the dual vector v_μ gives the second result.

²¹Proof: Let u and v be arbitrary tangent vectors. By invariance of the inner product: $g_{\mu\nu} u^\mu v^\nu = \langle \mathbf{u}, \mathbf{v} \rangle = g'_{\alpha\beta} u'^\alpha v'^\beta = g'_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} u^\mu v^\nu$. Since the vector components u^μ and v^ν are arbitrary, this implies: $g_{\mu\nu} = g'_{\alpha\beta} \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu}$.